Amazon Sales Data Analysis Report

1. Introduction

Amazon is one of the world's largest e-commerce platforms that offers products across multiple categories such as Electronics, Clothing, Beauty, Home, and Sports. With millions of daily transactions, analyzing sales data becomes important for understanding customer behavior, product performance, pricing trends, and business growth.

This project focuses on analyzing Amazon sales data using Python libraries like Pandas, NumPy, Matplotlib, and Seaborn. Different visualization techniques were used to identify trends, patterns, and relationships within the dataset.

The main goal of this analysis is to generate meaningful business insights related to sales trends, customer activity, ratings, pricing strategies, and payment methods.

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2. Objectives

The objectives of this project are:

- To analyze monthly sales trends and revenue performance.
- To study customer activity and spending behavior.
- To analyze category-wise product pricing.
- To examine customer ratings across different product categories.
- To understand payment method usage patterns.
- To identify relationships between numerical variables using correlation analysis.
- To generate business insights through data visualization.

Tools and Technologies Used

The following tools and libraries were used in this project:

Tool / Library	Purpose
Python	Programming Language
Pandas	Data manipulation and analysis
NumPy	Numerical operations

Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook	Project development environment

Dataset Description

The dataset contains information related to Amazon products and customer-related attributes.

Important Columns in Dataset

- Product Name
- Product Category
- Brand
- Price
- Discounted Price
- Customer Rating
- Rating Count
- Reviews
- Delivery Status
- Product Details

The dataset provides useful information for understanding product performance, pricing patterns, and customer satisfaction.

Data Preprocessing and Cleaning

Before analysis, the dataset was cleaned and processed carefully.

Steps Performed:

1. Imported dataset using Pandas.
2. Checked dataset dimensions using shape.
3. Viewed dataset structure and column information.
4. Identified missing values using isnull().
5. Removed duplicate records.
6. Handled null values in important columns.
7. Converted price columns into numeric datatype.
8. Cleaned categorical values.

9. Performed datatype conversions for better analysis.
10. Prepared data for visualization.

Importance of Data Cleaning

Data cleaning improves data quality and ensures accurate analysis results. Incorrect or missing values can negatively affect visualizations and insights.

Graphs and Visualizations Used

The following visualizations were used in the project:

- Countplot
- Histogram
- Bar Chart
- Boxplot
- Scatter Plot
- Heatmap

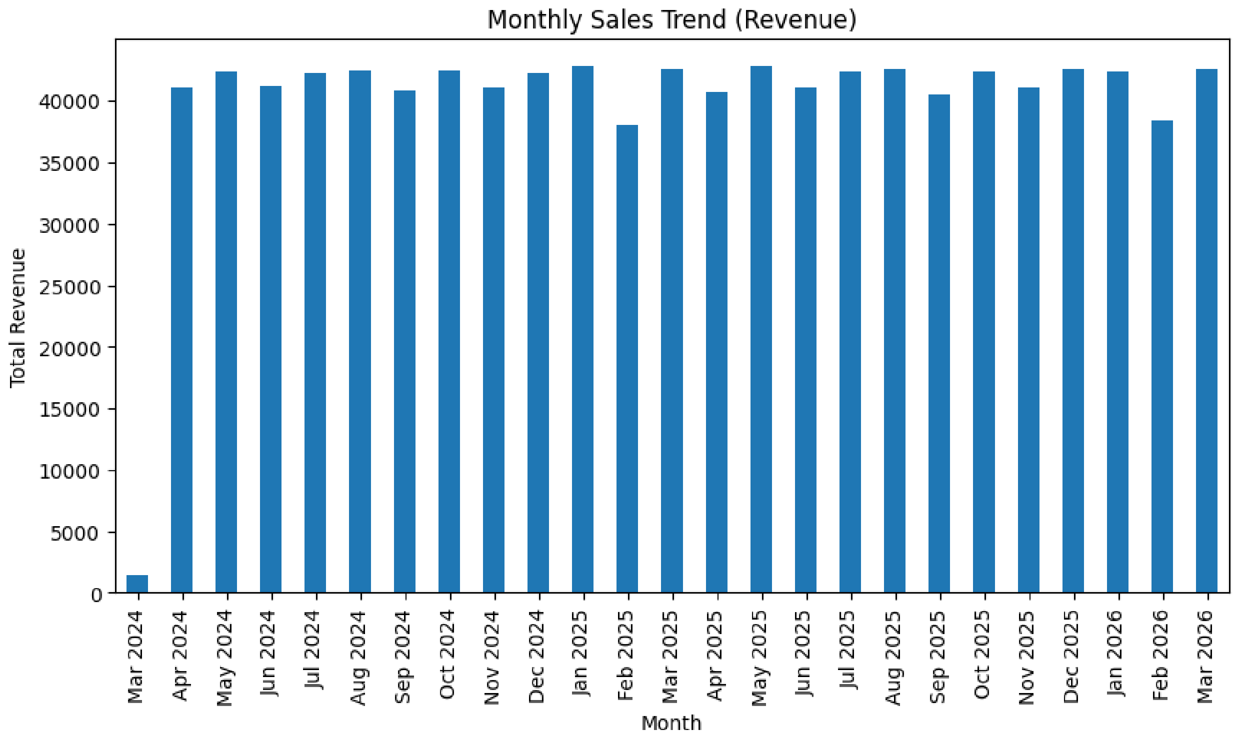
These visualizations improved understanding of sales trends, pricing patterns, and customer behavior.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand trends, patterns, and relationships in

3. Graph Analysis and Interpretation

3.1 Monthly Sales Trend (Revenue)



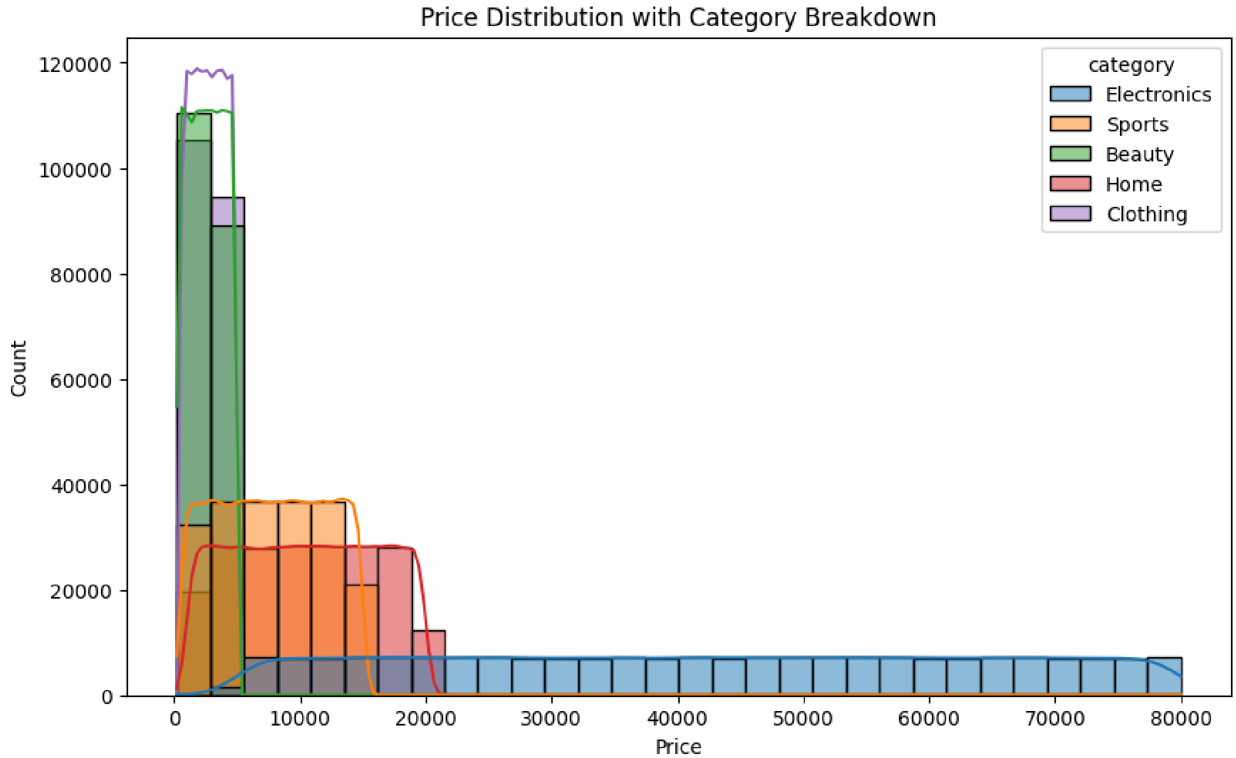
Observation

- Monthly revenue throughout the dataset period remained mostly stable and consistent.
- Most months generated revenue within a similar range, showing balanced business performance.
- A few months recorded slightly lower revenue, which may indicate seasonal demand fluctuations or reduced customer activity.
- Certain months achieved comparatively higher revenue, suggesting peak sales periods, promotional campaigns, or increased customer demand.
- Overall, the business maintained a healthy and steady revenue trend over time.

Insight

The analysis indicates that the company maintained stable sales performance across the observed period. Seasonal variations and promotional periods influenced revenue in certain months, but the overall trend reflects consistent business growth and customer engagement.

3.2 Price Distribution with Category Breakdown



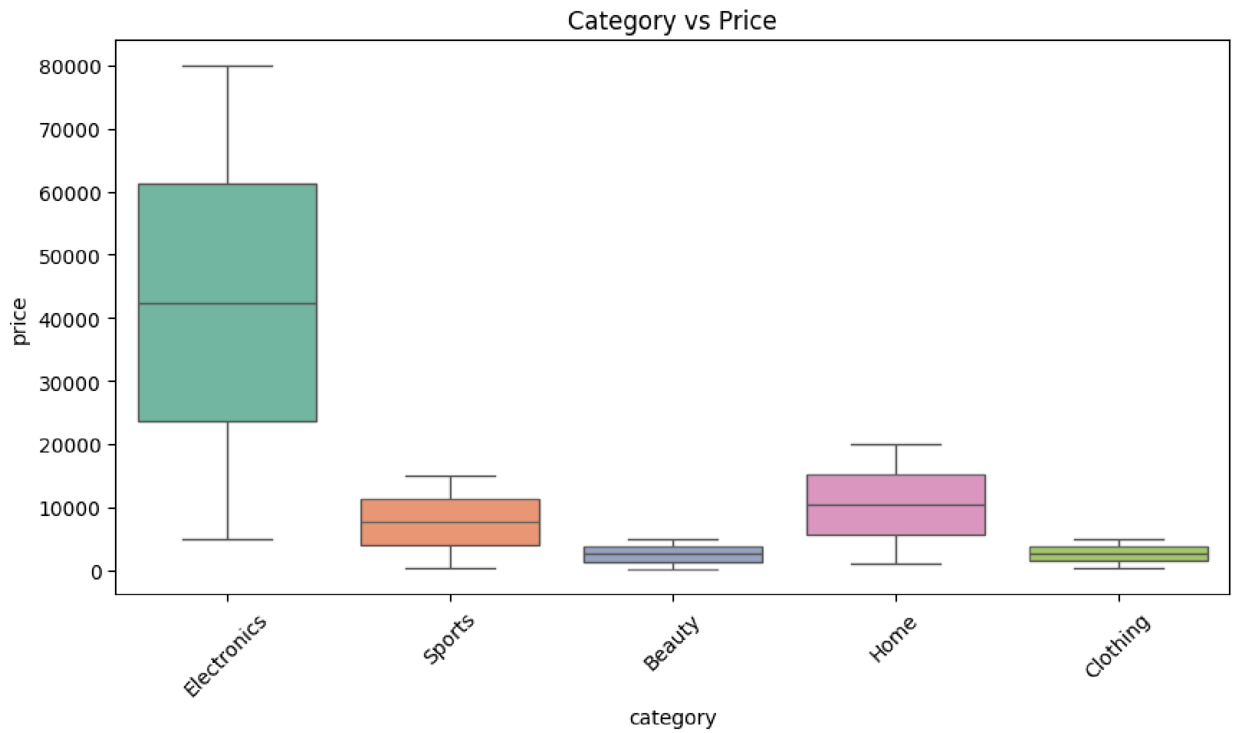
Observation

- Electronics category had the widest price range, extending to premium and high-value products.
- Home category products were mainly concentrated in the medium-price segment.
- Sports products showed moderate pricing distribution throughout the dataset.
- Beauty and Clothing categories mostly contained lower-priced and affordable products.
- Different product categories followed different pricing strategies based on target customers.

Insight

The pricing distribution analysis shows that customer purchasing behavior and pricing strategies vary significantly across categories. Premium categories target high-value customers, while affordable categories focus on mass-market demand.

3.3 Category vs Price Analysis (Boxplot)



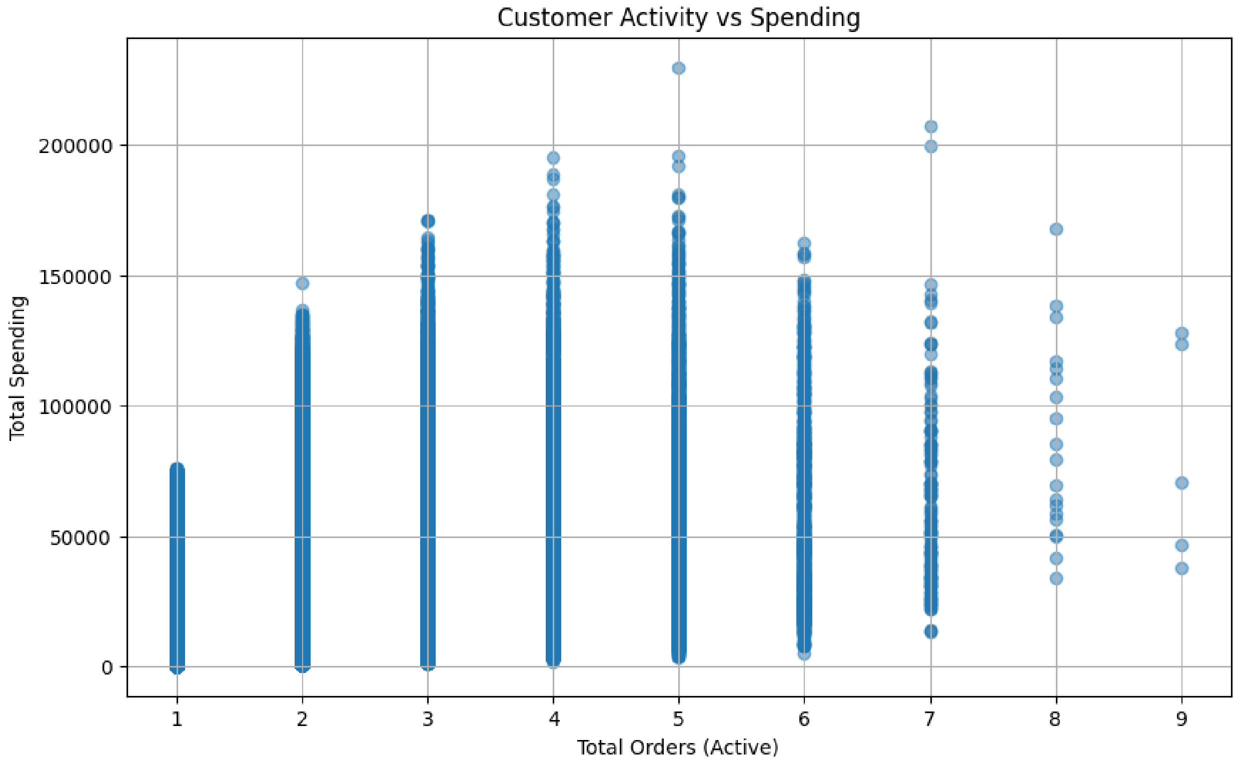
Observation

- Electronics products showed the highest price range among all categories.
- Beauty and Clothing products were comparatively lower priced.
- Home and Sports categories maintained moderate pricing levels.
- Electronics category displayed higher price variation, indicating both budget and premium products.

Insight

The boxplot analysis highlights category-wise pricing differences and helps identify categories with premium product positioning and greater price diversity.

3.4 Customer Activity vs Spend Analysis



Observation

- A positive relationship exists between customer activity and spending behavior.
- Customers placing more orders generally contributed higher overall spending.
- Repeat customers, especially those placing 4–7 orders, generated significant revenue.
- Some low-frequency customers also showed high spending behavior, indicating premium purchases.
- Customer retention strongly influenced total business revenue.

Insight

The analysis emphasizes the importance of repeat customers and customer retention strategies. Loyal customers contribute significantly to long-term revenue growth.

3.5 Rating vs Price Analysis (Scatter Plot)

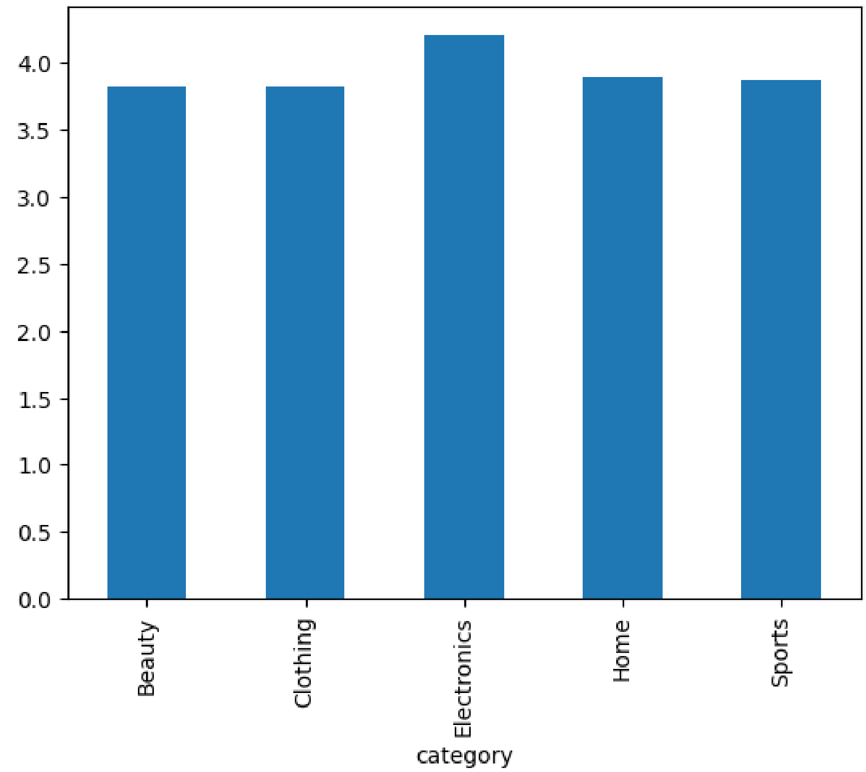
Observation

- Most product ratings ranged between 3 and 5.
- Products across different price ranges received similar customer ratings.
- High-priced products did not always receive higher ratings.
- Customer ratings remained mostly positive regardless of product price.

Insight

The analysis suggests that product quality and customer experience influence ratings more strongly than product pricing alone.

3.6 Average Rating per Category



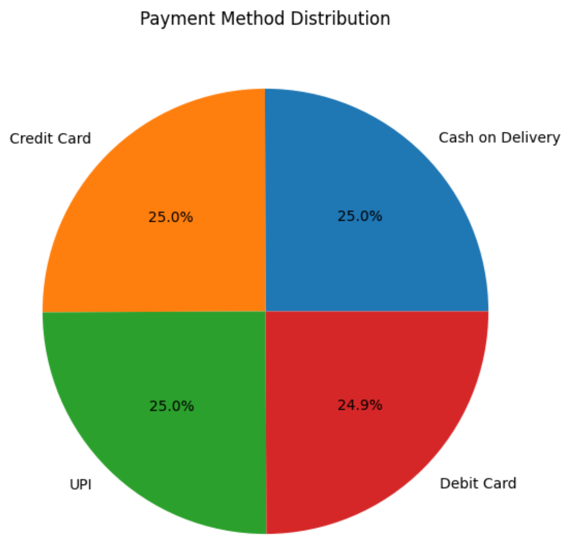
Observation

- Electronics category achieved the highest average ratings.
- Home and Sports categories also maintained positive customer satisfaction levels.
- Beauty and Clothing categories had slightly lower ratings but still remained within a positive range.
- Ratings across categories remained relatively balanced.
- Highly rated categories can be prioritized in recommendations and promotions.

Insight

Customer satisfaction remained strong across most categories, indicating overall positive product quality and user experience.

3.7 Payment Method Analysis (Pie Chart)



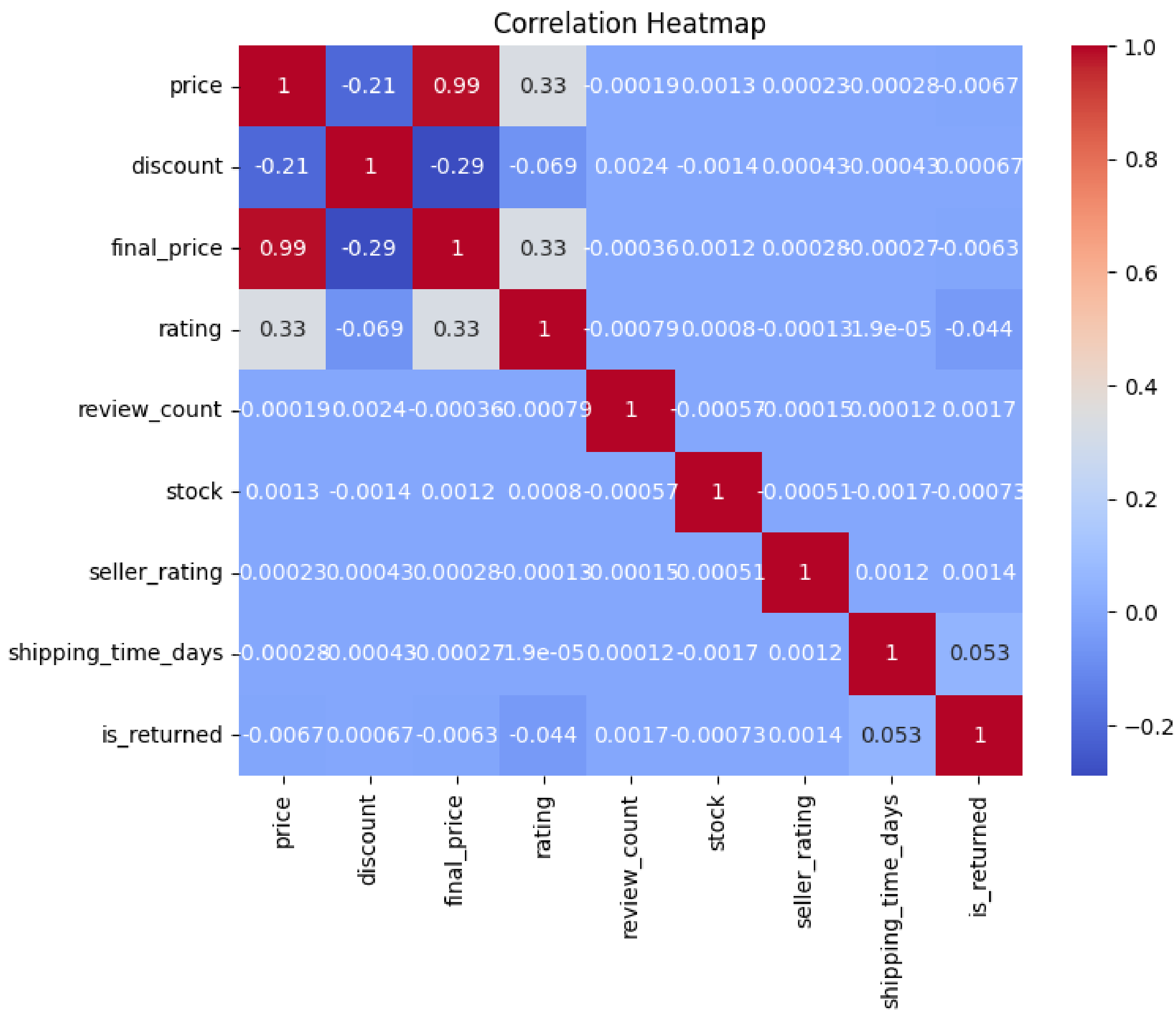
Observation

- Customers used multiple payment methods across the platform.
- Certain payment methods had larger usage shares, indicating higher customer preference.
- Digital payment methods contributed significantly due to faster and easier transactions.
- Balanced usage of payment methods reflects flexible customer purchasing behavior.
- Payment preferences influence customer convenience and checkout experience.

Insight

Understanding customer payment preferences helps businesses improve payment systems, transaction efficiency, and customer satisfaction.

3.8 Correlation Heatmap Analysis



Observation

- Strong positive correlation existed between price and final price variables.
- Review count and ratings showed moderate relationships, indicating customer trust influence.
- Discounts appeared to affect purchasing-related variables and customer engagement.
- Most variables did not show extremely strong correlations, suggesting multiple independent business factors.
- The heatmap helped identify relationships among numerical variables.

Insight

Correlation analysis provides valuable insights into how pricing, discounts, customer reviews, and engagement metrics influence business performance.

Graphs and Visualizations Used

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Key Findings

- Amazon contains products from multiple categories.
- Moderate-priced products are more common.
- Customer ratings are generally positive.
- Some categories contain premium-priced products.
- Product pricing varies significantly across categories.
- Successful deliveries dominate the dataset.
- Visualizations help identify important business insights effectively.

Advantages of the Project

- Helps understand customer behavior.
- Improves business decision-making.
- Identifies pricing trends.
- Supports product performance analysis.
- Demonstrates practical use of data analytics.

Conclusion

This project successfully analyzed Amazon e-commerce dataset using Python libraries and visualization techniques. Data cleaning and exploratory data analysis helped generate meaningful insights related to pricing, product categories, customer ratings, and delivery status.

The project demonstrates how data analytics techniques can be applied in e-commerce businesses for understanding customer preferences and improving decision-making.

Future Scope

Future improvements can include:

- Machine learning-based sales prediction.
- Product recommendation systems.
- Interactive dashboard development using Power BI or Tableau.
- Sentiment analysis of customer reviews.
- Customer segmentation analysis.
- Real-time sales monitoring systems.

Thank You